

STAT:5400 Fall 2026

Computing in Statistics

Lectures MW 5:00 PM –6:15 PM 75 SH

Instructor	Boxiang Wang, 362 SH, boxiang-wang@uiowa.edu
Office Hours	2:00 PM – 5:00 PM Monday
Teaching Assistant	Zhihan Chen, zhihan-chen@uiowa.edu
Department	Statistics & Actuarial Science, 241 SH. http://www.stat.uiowa.edu
Interim DEO	Professor Alberto Maria Segre, DEO, alberto-segre@uiowa.edu

Course Website.

Announcements, homework problems, etc: <http://icon.uiowa.edu>.

Lecture Notes.

Interactive lecture notes (code runs in your browser): <https://boxiang-wang.github.io/stat5400/>. A printable version is posted on ICON.

Course Schedule.

https://docs.google.com/spreadsheets/d/1nmDBqY4sLQ0DeLcnNHxXVw_gguzISeYp_jnDtXQT4t7Y/edit?usp=sharing

Course Description and Objectives.

This is a required course for master students in statistics and data science. Through hands-on experience with real problems, we will learn computing skills essential in applied statistics and in research in methodological and theoretical statistics.

This course is bilingual: coding is in R and Python.

The course mainly consists of two components.

- Component 1: Introduction to Statistical Computing Methods. Topics include generating random variables, Monte Carlo studies, resampling methods, computing linear models, computations in high-dimensional statistics, machine learning algorithms, and large language models.
- Component 2: Tech Guide. Topics include the Linux operating system, R Markdown, GitHub (version control), Sublime (text editor), C and Fortran, writing R packages, parallel computing, R packages ellmer, mall, localLLM, among other topics.

Course Home.

For undergraduate courses: The College of Liberal Arts and Sciences (CLAS) is the home

of this course, and CLAS governs the add and drop deadlines, academic misconduct policies, and other undergraduate policies and procedures. Other UI colleges may have different policies.

For graduate courses: The College of Liberal Arts and Sciences (CLAS) is the home of this course, and CLAS governs the policies and procedures for its courses. Graduate students, however, must adhere to the [academic deadlines set by the Graduate College](#).

Prerequisites.

CS:1210 and STAT:3200 and (STAT:3120 or STAT:3101 or STAT:4101).

Textbook.

No required textbook.

Optional Reference (in chronological order):

- Trevor Hastie, Robert Tibshirani, and Jerome H. Friedman (2009). *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. 2nd Edition, New York: Springer.
https://hastie.su.domains/ElemStatLearn/printings/ESLII_print12.pdf
- Norman Matloff (2011). *The Art of R Programming: A Tour of Statistical Software Design*. No Starch Press.
- Rafael A. Irizarry (2019). *Introduction to Data Science*. Chapman & Hall/CRC.
<https://rafalab.github.io/dsbook/>
- Maria L. Rizzo (2019). *Statistical Computing with R*. 2nd Edition, Chapman & Hall/CRC.
- Peter Bruce, Andrew Bruce, and Peter Gedeck (2020). *Practical Statistics for Data Scientists: 50+ Essential Concepts Using R and Python*. 2nd Edition, O'Reilly Media, Inc.
- Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani (2021). *An Introduction to Statistical Learning with Applications in R*. 2nd Edition, New York: Springer. (A Python edition, 2023, is also available.)
<https://www.statlearning.com/>
- Wes McKinney (2022). *Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter*. 3rd Edition, O'Reilly Media, Inc.
<https://wesmckinney.com/book/>
- Hadley Wickham, Mine Çetinkaya-Rundel, and Garrett Grolemund (2023). *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. 2nd Edition, O'Reilly Media, Inc.
<https://r4ds.hadley.nz/>

- Jay Alammam and Maarten Grootendorst (2024). *Hands-On Large Language Models: Language Understanding and Generation*. O'Reilly Media, Inc.

Grading.

Your semester grade will consist of the following components:

Homework	20%	
Midterm Exam	35%	Nov 2, 2026
Instructional Project	15%	
Computing Project	30%	
Total	100%	

As a **rough** guide, **A,A-** = 90% – 100%, **B+,B,B-** = 80% – 90%, **C+,C,C-** = 70% – 80%, **D+,D,D-** = 60% – 70%.

Grading will be on a curve, with +/– grade used. A grade of A+ represents exceptional work and is given only in extraordinary situations.

Homework.

In general, weekly homework will be assigned each Friday and will be due at 9:30 AM the following Friday. Exceptions to this schedule will be announced in class. Homework should be submitted electronically through the ICON submission tools for this course (<https://icon.uiowa.edu>). Show your work when solving written homework problems. Complete code and output must be submitted for computer problems. Always document your answer.

You are encouraged to study with others. However, if you work with others on homework, please: a) write up your own assignment and make sure you completely understand all solutions that you submit, and b) write the names of the others in your study group on your assignment.

For some homework questions, AI-generated text or code is prohibited. Details will be given in homework and will be discussed in class.

Late homework is accepted only as required by the university policy, i.e. due to “illness, mandatory religious obligations, or other unavoidable circumstances or University activities.” This is a strict rule; however, you are allowed to drop the lowest homework grade.

Office Hours.

2:00 PM – 5:00 PM Monday

TA office hour:

Zhihan Chen 10:00 AM - 11:00 AM, Thursday, 350 SH

Midterm Exam.

There is a two-hour hands-on computing midterm exam that will be held 5:00 PM – 7:00 PM on Nov 2, 2026 in 41 SH. The computing exams are open book and open notes. Web-search or AI is prohibited. You are not allowed to discuss with others about the exam.

Students must take exams as scheduled except in cases of officially university-approved absence such as class conflict with official exam time, illness, religious observance, and NCAA athletic competition.

Makeup exams are not available for other reasons, including student org field trips, club competitions, job interviews, and personal events.

More teaching policies and resources are seen in:

<https://clas.uiowa.edu/faculty/teaching-policies-resources-examination-policies>.

Instructional Project.

Students will work in groups of three to study and present a topic that is not covered in this class. You are expected to give a whole lecture to introduce the topic to other students. You are free to use any resource, including some tutorials available online, but you must cite them clearly. You are encouraged to design your lecture in the format of a lab, allowing students to have hands-on experience of the tools that you introduce. You are also required to make a short homework question based on your material.

Possible topics:

1. `dplyr` & the `tidyverse` – data wrangling in R
2. `ggplot2` – publication-quality graphics
3. `data.table` vs. `polars` – high-performance data manipulation
4. `shiny` – interactive web applications in R
5. Web scraping – harvesting data from websites with `rvest`/`httr2`
6. SQL with DuckDB – Text2SQL (AI-driven)
7. Regular expressions & text processing
8. `gt` – publication-quality tables
9. `plotly` and `gganimate` – interactive & animated graphics
10. Tableau or Power BI
11. Julia
12. SAS
13. Apache Arrow & Apache Parquet – analyzing larger-than-memory data on a laptop

14. **targets** – reproducible analysis pipelines in R
15. Docker – containerizing an analysis so it runs anywhere
16. NotebookLM – source-grounded AI notebooks
17. AI media generation – Seedance, Sora, Veo, and image models
18. AI research assistants – Elicit, Perplexity, and deep-research tools for literature work

The project is collaborative. Each student on a team is expected to complete a similar amount of work and to contribute equally to the final project. Students who misrepresent themselves as equal partners in this collaborative project but who are actually letting others do the bulk of the work will be reported to the College for academic dishonesty.

Computing Project.

Students will work in groups of four to carry out projects involving application of the statistical computing methods covered in the course to problems of their own choosing. Examples of possible types of projects are:

- Design and carry out a simulation study to compare the properties of two or more statistical procedures.
- Choose a research paper. Use simulation and find data to verify results in the paper.
- A real-world project that will be provided later by the instructor.

The computing project will be carried out in two stages.

- Project proposal (due 5:00 PM, Nov 9, 2026. The grade is reduced by 30% if the submission is late by less than one day; no credit is given after that.)

Each team should write a one-page summary. This is a detailed description of what you plan to do, including questions to be addressed, software to be used, and methods to be applied.

No proposal is needed for the team who plans on a real-world project.

- Project presentation.

Each team will give a 35-minute presentation on Dec 2, Dec 7, or Dec 9, 2026. A list of which tasks each team member has performed should be submitted.

It is required to work with different people for the instructional and computing projects.

Copyright.

Course materials that the instructor has posted publicly, for example, the course lecture-notes website, may be shared and used freely. All other course materials, including homework, quizzes, exam questions, solutions, and anything distributed through ICON, are for use only by students enrolled in STAT:5400 and may not be shared with anyone

else without the instructor's permission.

The unauthorized video or audio recording of academic activities (e.g., lectures, course discussions, office hours, etc.) by a student is prohibited. Students with a reasonable accommodation for recording approved by Student Disability Services should notify each instructor and provide the Letter of Accommodation prior to using the accommodation. A student may record classroom activities with prior written permission from the instructor and notice to other students in the class that audio or video recording may occur. Any and all classroom recording must be for personal academic use only. The distribution, sharing, sale, or posting of recordings on the internet (including social media), in whole or in part, is prohibited and doing so may be a violation of the Code of Student Life and/or state or federal privacy, copyright, or other laws.

Artificial Intelligence (AI) Policy

This course teaches statistical computing with AI tools, and their use is allowed and expected by default. Exceptions are explicitly marked: homework problems and code blocks labeled “by hand” must be completed without AI assistance, and the midterm exam is entirely AI-free. For AI-assisted work, you remain fully responsible for correctness — verify all results by running code, and be prepared to explain any line you submit.

Attendance.

Attendance is required. See instruction for Absence from Class due to illness, emergency, or extenuating circumstances in <https://registrar.uiowa.edu/absence-class> .

A student who is absent for more than five days or miss an exam due to an emergency or an illness may request the UI Service Center, 17 Calvin Hall, registrar@uiowa.edu, to notify me of the reason for the absence.

“Incomplete” Grades A grade of “Incomplete” will only be given under extraordinarily extenuating circumstances that prevent the student from completing the course requirements. Having a failing grade in the course is not an extraordinarily extenuating circumstance.

Academic Honesty and Misconduct

All students in CLAS courses are expected to abide by [the college's standards of academic honesty](#). Undergraduate academic misconduct must be reported by instructors to CLAS according to these procedures. Graduate academic misconduct must be reported to the Graduate College according to Section F of the Graduate College Manual.

Student Complaints

Students with a complaint about a grade or a related matter should first discuss the situation with the instructor and/or the course supervisor (if applicable), and finally with the DEO (Chair) of the department, school or program offering the course. Sometimes students will be referred to the department or program's Director of Undergraduate

Studies (DUS) or Director of Graduate Studies (DGS).

Undergraduate students should contact [CLAS Undergraduate Programs](#) for support when the matter is not resolved at the previous level. Graduate students should contact the [CLAS Graduate Affairs Manager](#) when additional support is needed.

Drop Deadline for this Course

You may drop an individual course before the drop deadline; after this deadline you will need collegiate approval. You can look up the [drop deadline for this course](#) here. When you drop a course, a “W” will appear on your transcript. The mark of “W” is a neutral mark that does not affect your GPA. To discuss how dropping (or staying in) a course might affect your academic goals, please contact your Academic Advisor. Directions for adding or dropping a course and other registration changes can be found on the [Registrar’s website](#). Undergraduate students can find policies on dropping CLAS courses [here](#). Graduate students should adhere to the [academic deadlines](#) and policies set by the Graduate College.

Communication: UI Email

Students are responsible for all official correspondences sent to their UI email address (uiowa.edu) and must use this address for any communication with instructors or staff in the UI community. For the privacy and the protection of student records, UI faculty and staff can only correspond with UI email addresses.

Where to Get Academic Support for This Course

Office hours (Monday 2:00–5:00 PM, 362 SH), the teaching assistant, and the ICON discussion board are the first stops for help with this course. Questions about the lecture notes or homework are welcome at any time by email.

Mental Health Resources and Student Support

Students are encouraged to be mindful of their mental health and seek help as a preventive measure or if feeling overwhelmed and/or struggling to meet course expectations. Students are encouraged to talk to their instructor for assistance with course-related concerns. For additional mental health support, please see the guidance and resources at mentalhealth.uiowa.edu, including the 24-7 [UI Support and Crisis Line](#).

Additionally, the Office of the Dean of Students can help students navigate personal crisis situations. They can provide one-on-one support, help with identifying options, and access to [basic needs resources](#) (such as food, rent, childcare, etc.). Student Care and Assistance: 132 IMU, dos-assistance@uiowa.edu, or 319-335-1162 and more info: dos.uiowa.edu/assistance.

[Accommodations for Students with Disabilities](#)

The University is committed to providing an educational experience that is accessible to all. If a student has a diagnosed disability or other disabling condition that may impact

the student's ability to complete the course requirements as stated in the syllabus, the student may seek accommodations through [Student Disability Services](#) (SDS). SDS is responsible for making [Letters of Accommodation \(LOA\)](#) available. The student must provide an LOA to the instructor as early in the semester as possible, but requests not made at least two weeks prior to the scheduled activity for which an accommodation is sought may not be accommodated. The LOA will specify what reasonable course accommodations the student is eligible for and those the instructor should provide. Additional information can be found on the SDS website.

University Policies

[Basic Needs and Student Support](#)

[Classroom Expectations](#)

[Absences from Class](#)

[Free Speech and Expression](#)

[Mental Health](#)

[Absences for Military Service Obligations](#)

[Non-discrimination Statement](#)

[Absences for Religious Holy Days](#)

[Sexual Harassment/Sexual Misconduct and Supportive Measures](#)

[Class Recordings](#)

[Conflict Resolution](#)